

Computer Vision for CS & AI

SESSION 1 · IMAGES AS NUMBERS

From pixels to representations — a two-hour interactive lecture

Mohammad Sabokrou · New Uzbekistan University

00-03 MIN

What can machines see today?

Computer vision is no longer only “classify this photo.”

It measures, locates, tracks, reconstructs, generates, and supports decisions.

Opening question: Which application would you trust with your life?

APPLICATION 1

Autonomous driving



See lanes, vehicles, pedestrians, lights, and free space — continuously.

Core tasks · detection · depth · tracking · segmentation

Hard case · rain, glare, unusual roads, and a child partly hidden by a car

APPLICATION 2



Medical imaging

Highlight suspicious regions in X-rays, CT, MRI, pathology, and retinal images.

Vision supports clinicians; it does not remove clinical judgement.

Hard case · rare disease, scanner shift, demographic bias, and false reassurance

APPLICATION 3

Earth observation & disaster response



Compare satellite images to map floods, wildfire damage, deforestation, and damaged buildings.

The problem is often change detection: what changed, where, and how certain are we?

Hard case · clouds, seasons, registration errors, and limited labels

APPLICATION 4



Precision agriculture

Detect crop stress, weeds, pests, fruit, and irrigation problems from drones or phones.

A useful system must work across farms, seasons, cameras, and lighting.

Hard case · learning the soil or background instead of the disease

APPLICATION 5



Manufacturing inspection

Find scratches, cracks, missing components, contamination, and assembly errors.

Often we have many normal examples but few examples of every possible defect.

Hard case · anomaly detection under new products and changing production lines

APPLICATION 6

Sports analytics



Track players and the ball, estimate pose, recognize actions, and build tactical statistics.

One video produces a sequence of linked decisions, not independent images.

Hard case · occlusion, fast motion, camera cuts, and identity switches

APPLICATION 7



Augmented reality & spatial computing

Estimate camera pose, understand surfaces, anchor virtual objects, and recognize gestures.

The system must connect 2D pixels to a stable 3D world in real time.

Hard case · reflective surfaces, low texture, motion blur, and latency

APPLICATION 8

Generative and multimodal vision



Generate and edit images, answer scene questions, retrieve images with language, and reason across video.

Generation still depends on representations and may reproduce data errors or biases.

Hard case · hallucinated details, provenance, copyright, and manipulated media

03–20 MIN

The common structure

SCENE → PIXELS → REPRESENTATION → TASK OUTPUT → DECISION

What information should remain? What variation should be ignored? How do we know the system is reliable?

Where the course goes

PART I · Images, light, colour, histograms

PART II · Filtering, edges, features, geometry

PART III · Learning, neural networks, CNNs, transfer learning

PART IV · Detection, segmentation, tracking, evaluation and trustworthy systems

Every week: concept → prediction → implementation → failure analysis → evidence.

Syllabus · Weeks 1–8

- 01 Images as numbers; the recognition problem
- 02 Image formation, colour, intensity and histograms
- 03 Filtering, convolution and noise
- 04 Gradients, edges and corners
- 05 Local features and image matching
- 06 Geometry, transformations and alignment
- 07 Machine-learning foundations for vision
- 08 Neural networks and convolutional networks

Syllabus · Weeks 9–16

- 09 Training CNNs, transfer learning and augmentation
- 10 Robust evaluation and distribution shift
- 11 Object detection
- 12 Semantic and instance segmentation
- 13 Video, motion and tracking
- 14 Project development and experimental design
- 15 Trustworthiness: robustness, fairness and explainability
- 16 Project presentations, synthesis and review

How the course works

LECTURE · 2 hours

Concepts, worked examples, questions, and short prediction tasks

LABORATORY · 2 hours

Run an experiment, break a method, repair it, and record evidence

WEEKLY RECORD

An executed notebook with outputs, measurements, and a short reflection

SCORING

Assessment at a glance

10% Lab portfolio · Labs 1–15; Lab 0 is compulsory setup

15% Project work · developed through the project laboratories

75% Remaining assessment · follows the official module plan

If the official module plan changes, the official published plan takes precedence.

How each laboratory is scored · 20 marks

6 marks Correct, complete implementation that runs from first cell to last

4 marks Requested evidence: visible outputs, plots, tables, or measurements

4 marks Challenge task and deliberate failure case

3 marks Interpretation: explain what the evidence means

2 marks Reproducibility: clear order, fixed seeds where needed, saved outputs

1 mark Clear attribution of external or AI-assisted code

A notebook that does not run cannot receive implementation marks.

EXPECTATIONS

Class policy

ATTEND & PARTICIPATE · arrive ready to predict, discuss, and test

COLLABORATE · discuss ideas and debug together; submit your own work

ACADEMIC INTEGRITY · copied notebooks or unexplained code are not evidence of learning

AI TOOLS · mark assisted code with # ASSISTED and be ready to explain and modify it

DATA RESPONSIBILITY · do not upload private, sensitive, or unlicensed data

RESPECT · critique ideas and evidence, never people

Submission policy

Submit the executed notebook and short report by email to:

[email will be released]

Deadline · 23:59 on the day of your scheduled laboratory, unless announced otherwise

File names · StudentID_WeekXX.ipynb and StudentID_WeekXX_Report.pdf

Before sending · Runtime → Restart and run all; keep required outputs visible

Email subject · CV Lab XX — Student ID — Full name

Late or alternative arrangements must be agreed before the deadline.

Today's learning outcomes

By the end of Session 1, you should be able to:

- 1 Describe a grayscale or colour image as a numerical array
- 2 Inspect shape, data type, and value range before processing
- 3 Calculate a pixel distance and explain its hidden assumption
- 4 Identify nuisance variations that break raw-pixel comparison
- 5 Explain why computer vision needs useful representations
- 6 Separate signal, nuisance, and evaluation groups in a dataset

THE QUESTION

You knew before you calculated anything

A person sees an object almost immediately.

A computer receives only a grid of numbers.

Think–pair–share: What information did your brain use — colour, boundary, texture, context, memory, or all of them?

What an image is

GRAYSCALE · one grid $I(r,c)$ with shape (H, W)

COLOUR · three stacked grids with shape $(H, W, 3)$

8-BIT IMAGE · each channel value is usually an integer from 0 to 255

FLOAT IMAGE · values may instead be scaled to $[0,1]$ or another range

A 300×451 RGB image stores 405,900 channel values.

A 12-megapixel RGB photo stores roughly 36 million channel values.

WORKED EXAMPLE

A pixel has a location and channel values

Red R = [[240,220],[35,20]]

Green G = [[80,70],[190,170]]

Blue B = [[40,35],[60,55]]

Pixel at row 0, column 0 is RGB(240,80,40): bright red-orange.

Always inspect: shape · dtype · minimum · maximum

Also check: channel order · alpha channel · unexpected scaling

CHECKPOINT

Example: brightness is arithmetic

Original grayscale patch

$[[40, 80], [120, 200]]$

Add 40 $\rightarrow [[80, 120], [160, 240]]$

Multiply by 1.5 $\rightarrow [[60, 120], [180, 300]]$

But 8-bit images cannot store 300. Clip to 255 or convert before arithmetic.

Prediction: What happens if uint8 arithmetic wraps instead of clipping?

COMMON BUG

Example: RGB and BGR

The stored triplet $[255, 0, 0]$ can mean red or blue depending on channel convention.

RGB · used by many plotting and deep-learning tools

BGR · traditionally returned by OpenCV

Debugging rule: if skin, sky, or familiar objects have strange colours, verify channel order before changing the model.

The first proposal

“Store one example of each object and compare a new image pixel by pixel.”

- 1 STORE one template per object
- 2 COMPARE corresponding pixels
- 3 DECIDE: the nearest template wins

A sensible baseline. Its failure teaches us what a representation must accomplish.

WORKED EXAMPLE

Mean absolute difference · step by step

Template $T = [[10,20],[30,40]]$

Query $Q = [[12,18],[33,37]]$

$|Q - T| = [[2,2],[3,3]]$

$$\text{MAD} = (2 + 2 + 3 + 3) / 4 = 2.5$$

The calculation is correct. The fragile assumption: corresponding pixels describe corresponding scene parts.

2-MINUTE TASK

Try it before revealing the answer

$$A = [[0,0],[255,255]]$$

$$B = [[0,10],[245,255]]$$

$$C = [[255,255],[0,0]]$$

- 1 Calculate $MAD(A,B)$
- 2 Calculate $MAD(A,C)$
- 3 Which is closer to A?
- 4 Does “closer pixels” always mean “same object”?

Answer: $MAD(A,B)=5$; $MAD(A,C)=255$.

Why recognition is hard

ILLUMINATION · same surface, different values

VIEWPOINT · visible shape and arrangement change

SCALE · object occupies a different number of pixels

OCCLUSION · part of the evidence disappears

SENSOR NOISE · recorded values vary

DEFORMATION · object geometry changes

BACKGROUND CLUTTER · irrelevant structure competes

INTRA-CLASS VARIATION · different examples share one label

A one-pixel shift

Original [0,0,255,255,0,0]

Shifted [0,0,0,255,255,0]

Four positions agree; two differ by 255.

$$\text{MAD} = (255 + 255) / 6 = 85.$$

The object did not change, but pointwise correspondence did.

Question: Would a histogram notice this shift?

Predict the failure

Two changes are applied to the same cup image:

- A The entire image becomes darker
- B Twenty percent of the cup is covered

Which produces the larger raw-pixel distance?

Vote first. Give one reason. Then examine the measurement.

45-60 MIN

The surprising measurement

Dimming illumination can change more pixels than covering part of the object.

A raw-pixel rule may call a dark cup less similar than a heavily occluded cup.

Observation · lighting strongly changes recorded values

Conclusion · raw pixel distance does not reliably measure identity

Next · what preserves identity while reducing lighting sensitivity?

60–65 MIN

Break · five minutes

When we return:

Why memorisation is impossible, what a representation is, and how dataset design changes what a model learns.

Why not store every possible image?

A 224×224 RGB image contains 150,528 values.

With 256 choices per value:

$$256^{150,528} \approx 10^{362,000}$$

A working system must generalize using structure; exhaustive memorisation is arithmetically impossible.

Representation: keep signal, discard nuisance

A representation preserves information needed for a task.

Examples · histogram · edge map · local descriptor · CNN feature vector

For object identity, it should keep shape or semantic evidence while reducing irrelevant lighting or small shifts.

There is no universally best representation; the task defines what should remain invariant.

Compression can preserve recognition

$224 \times 224 \times 3 = 150,528$ values

$16 \times 16 \times 3 = 768$ values

The smaller image uses about $196\times$ fewer values.

Recognition can survive because neighbouring pixels repeat structure.

Discussion: Which disappears first — texture, exact boundary, object shape, or identity?

The data is made, not found

One scene with three exposures produces three different arrays and histograms.

Images inherit choices:

camera · lighting · viewpoint · compression · operator · place · time · sampling

A model trained only on one condition may treat another valid condition as unfamiliar.

Signal, nuisance, and shortcut

SIGNAL · what should determine the answer

Example: disease evidence in a medical image

NUISANCE · variation that should not change the answer

Example: scanner model or exposure

SHORTCUT · an easier accidental predictor

Example: hospital marker appears mostly in positive training images

A high test score can still reflect the shortcut if the split repeats it.

Design the dataset before training

- 1 What exactly should determine the label?
- 2 Which nuisance factors must vary within every label?
- 3 Which groups or conditions need separate evaluation?
- 4 What must not leak across train and test?

Keep near-duplicates and frames from the same capture sequence in only one split.

An overall score can hide failure

Group A · 2 / 5 correct = 40%

Group B · 0 / 5 correct = 0%

Overall · 2 / 10 correct = 20%

Reporting only 20% conceals complete failure for Group B.

Course rule: report meaningful group results and failure cases, not only a single average.

88–105 MIN

Mini design challenge

Choose one opening application. In groups of 3, write:

- 1 INPUT — what image or video enters?
- 2 OUTPUT — what must it predict?
- 3 SIGNAL — what should determine the prediction?
- 4 NUISANCES — name at least three
- 5 FAILURE GROUPS — where might an average hide harm?
- 6 EVIDENCE — what measurement would convince you?

Prepare a 45-second explanation.

Session summary

- 1 An image is an array; shape, dtype, range, and channel order matter
- 2 Pixel distance assumes aligned correspondence
- 3 Nuisance changes can dominate raw-pixel similarity
- 4 A representation preserves task signal while reducing nuisance sensitivity
- 5 Dataset design determines available shortcuts
- 6 Evaluation needs groups and failure cases, not only an average

Before Lab 1 · complete Lab 0

- ✓ Open the notebook and save a copy to Drive
- ✓ Run every cell in order
- ✓ Load and inspect the supplied dataset
- ✓ Read one image and write one image
- ✓ Save results and read them back
- ✓ Finish with a green environment report

Lab 0 is compulsory setup work. It is not scored for correctness.

Self-study check

Explain without looking back:

- 1 Which four properties should you print after loading an image?
- 2 Why can a one-pixel shift produce a large distance?
- 3 What assumption makes template matching fragile?
- 4 Give one signal and two nuisances for face recognition
- 5 Why can overall accuracy conceal serious failure?
- 6 What is a representation, in your own words?

Next week

You will build the simple classifier,
watch it succeed,
break it with illumination,
and repair it with your first representation.

Questions? · 118–120 min